

Task Manager API with PDF Export

Role: Software Developer

Difficulty: junior

Project Type: fullstack

Duration: 3 days

Focus: api_development

Candidate: Rahul Sharma

Project Overview

Build a task API with JWT auth and PDF export, plus a Next.js UI.

Technologies

- JavaScript
- TypeScript
- React
- Next.js
- Node.js
- Express.js
- Laravel (PHP)
- Tailwind CSS
- MongoDB
- JWT

Requirements

- Use existing skills, add new learning, measurable outcome

Objectives

- Implement JWT auth
- Create CRUD endpoints
- Build Next.js pages
- Export tasks to PDF
- Document API

Functional Requirements

- POST /auth/login
- GET /tasks
- POST /tasks
- PUT /tasks/:id
- DELETE /tasks/:id
- GET /tasks/pdf

Non-Functional Requirements

- JWT security
- Input validation
- Error handling
- Responsive UI
- Performance <200ms

Modules

Auth

JWT login

- login
- token

Tasks

CRUD

- list
- create
- update
- delete

PDF

Export

- generate
- download

UI

Next.js pages

- list
- form
- pdf

API

Express routes

- routes
- middleware

DB

MongoDB models

- schema
- connection

Implementation Plan

Setup

- init repo
- install deps

Auth

- create login route
- JWT middleware

Tasks

- model
- routes
- validation

PDF

- pdf lib
- endpoint

UI

- pages
- api calls

Testing

- unit tests
- integration tests

Evaluation Criteria

Auth (20%)

JWT works

CRUD (20%)

Endpoints functional

PDF (15%)

PDF generated

UI (15%)

Responsive design

Docs (10%)

API docs present

Tests (20%)

Coverage >70%

Deliverables

- Git repo
- API docs
- PDF sample
- UI screenshot

Suggested Folder Structure

- components
- pages
- api
- models
- routes
- utils
- public
- styles

API Endpoints

POST /auth/login

login

GET /tasks

list

POST /tasks

create

PUT /tasks/:id

update

DELETE /tasks/:id

delete

GET /tasks/pdf

export pdf

Database Design

users

auth

- email
- passwordHash

tasks

tasks

- title
- description
- status
- ownerId

Testing Plan

- unit tests for routes
- integration tests for auth
- e2e test for PDF

Candidate Fit

Matches Rahul's skills, adds JWT and PDF learning